

Project Ideas



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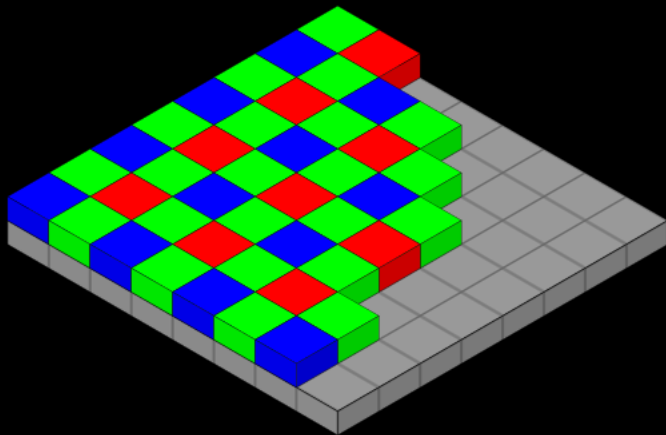


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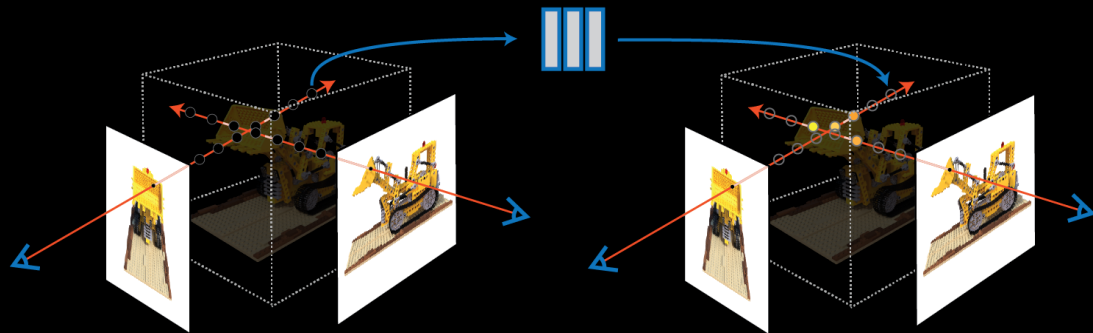
List of Projects

1. RGB Demosaicking + neural radial Field (NERF)
2. 3D point Cloud and Gaussian splatting
3. Event based and Gaussian Splatting
4. Single pixel camera + Gaussian splatting
5. Snapshot video imaging
6. Multispectral demosaicking + filter design
7. Multispectral deconvolution

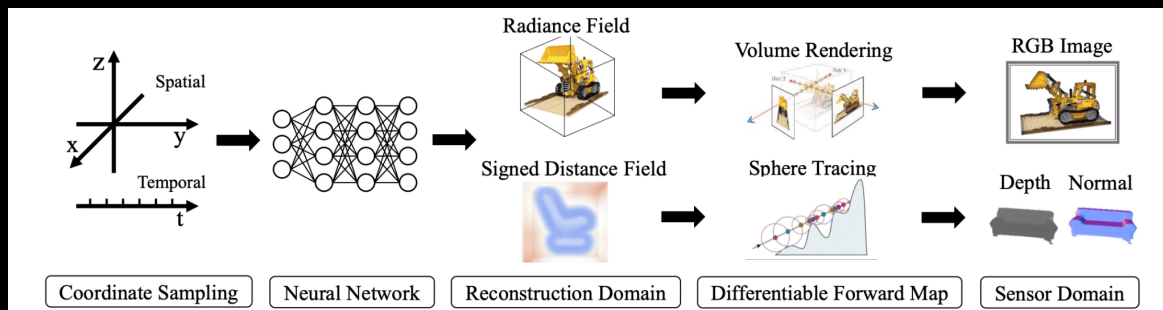
RGB Demosaicking + Neural Radial Fields



Bayer filter



Neural Radial Fields (NERF)



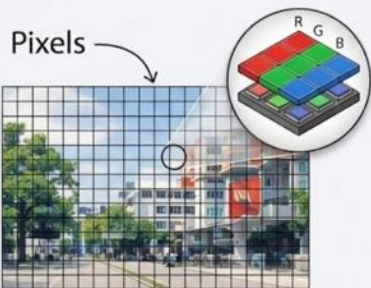
NERF requires
multiple view angles

Point Cloud + Gaussian Splatting



Event Camera+ Gaussian Splatting

RGB Camera



30 Samples entire scene at regular intervals (e.g., 30 fps)

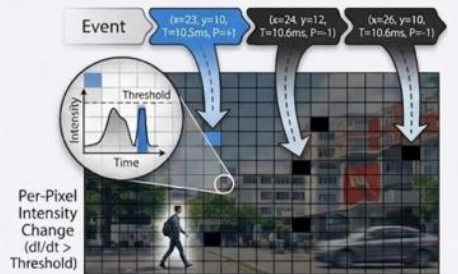
Processing & Frame Generation



Output Frame at T1 (Static, Redundant Data)

Advantage: Rich color, dense data

Event Camera



Continuous, Asynchronous Data Stream

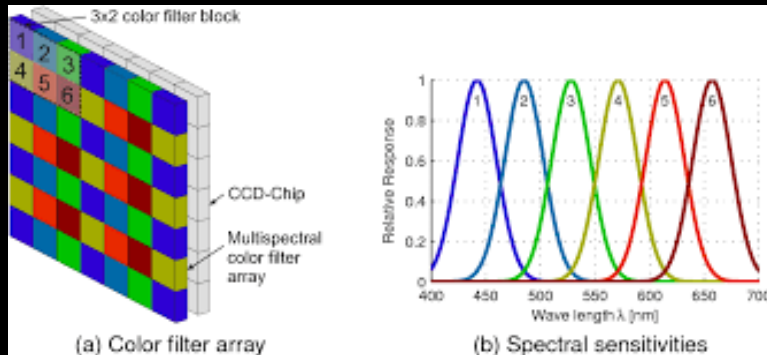


Event Accumulation & Trail (Sparse, Novelty Data)

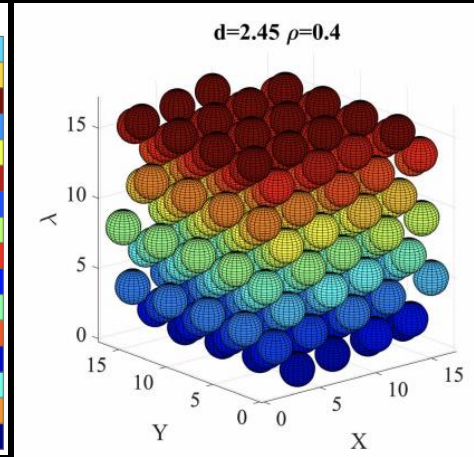
Advantage: High speed, low power, data-sparse



Multispectral Filter Array + Sphere Packing Bilinear Demosaicking for any Number of Filter



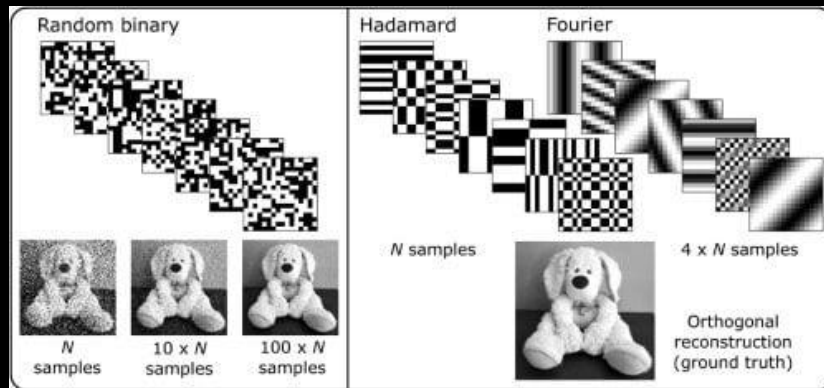
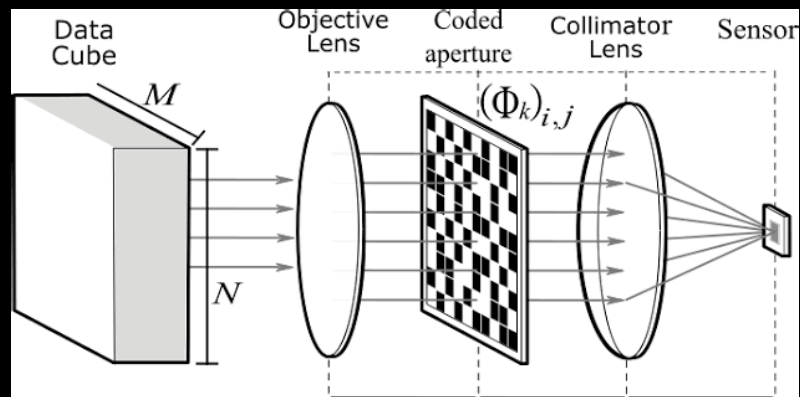
Multispectral Filter Array



Sphere packing
Multispectral Filter Array

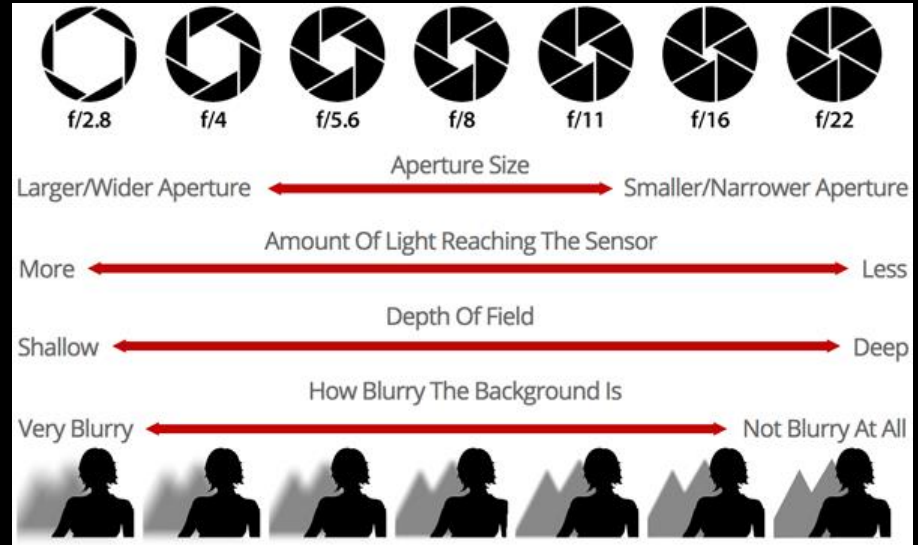
Goal: To find the deconvolution kernel for any number of Filter?

Single Pixel Camera + Gaussian splatting



Depth-of-Field (DoF)

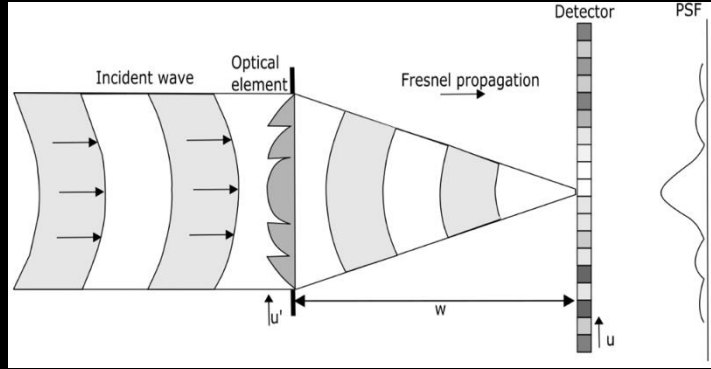
The **depth-of-the-field** refers to the depth range of an object that can be **sharply imaged** by a given optical imaging system and is determined by **operating wavelength**, **effective focal length** and **aperture size** of the imaging lens.



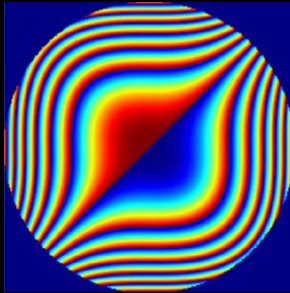
- Reducing the aperture size improves the depth of field
- But reduces the incoming light

Extended-DoF with Diffractive Optical Element

Two assumption when designing EDoF

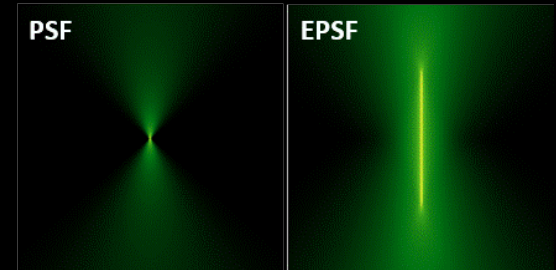


2. It is assumed that any resulting image will be an **intermediate image** and will require **digital processing**¹



1. The **incoherent optical system** is being modified by a rectangular separable **phase mask**¹

Standard vs Extended PSF



- Conventional cameras has narrow PSF
- The idea of EDoF is to promote PSF insensitive to misfocus

1. E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).

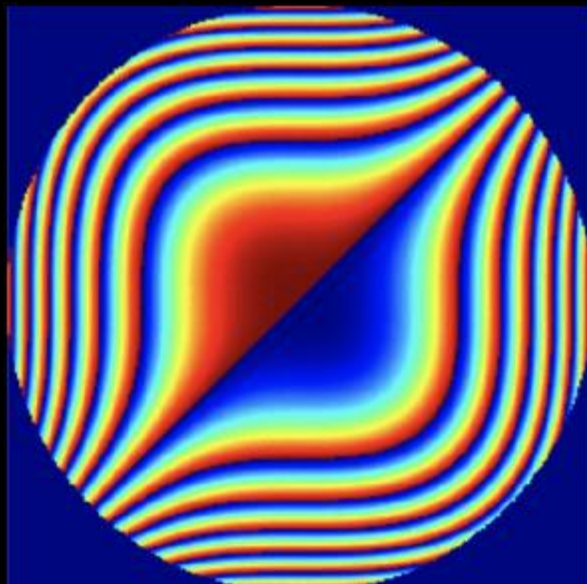
EDoF Discrete Model

$$Y_{RGB,z} = S_{RGB}(\textcolor{red}{PSF}_{\lambda,z} \odot X_{\lambda,z}) + \Omega_{\lambda,z}$$

designing the encoding is critical
to facilitate the inverse problem



measurement



DOE



Scene